

Pavo Cloud

Autonomous Prior Authorization

The insurer's answer in **88.6 milliseconds**. The whole round trip in under five minutes, not three to fourteen days.

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Prior Authorization Is Broken

Before many procedures, the insurer has to agree to pay. In practice that is a member of staff filling in a form, faxing it, and then waiting.

3–14 days

to a decision

94%

of doctors report delays

97%

of requests approved in the end

\$35B

spent on the process each year

Most of that \$35 billion buys nothing. The great majority of requests are approved anyway, just late.

The Three Eras of Prior Authorization

ERA 1

Fax and Phone

A person starts it and a person resolves it.

~4 weeks

ERA 2

Digital Portals

A person starts it, the portal assists.

3-14 days

ERA 3

Autonomous Agents

Software starts it and software resolves it.

Under 5 min

Only the third era removes the person from the routine cases, so clinicians spend their time on the hard ones.

Era 3: Agent to Agent

Submitted automatically

Clinical documentation leaves the record system directly. No staff touchpoint, no portal login.

Clear cases close in milliseconds

The full decision path takes **88.6 milliseconds**, and 84.0 of those are the two RSA-2048 signatures every authorization writes. The deciding itself is the cheap part.

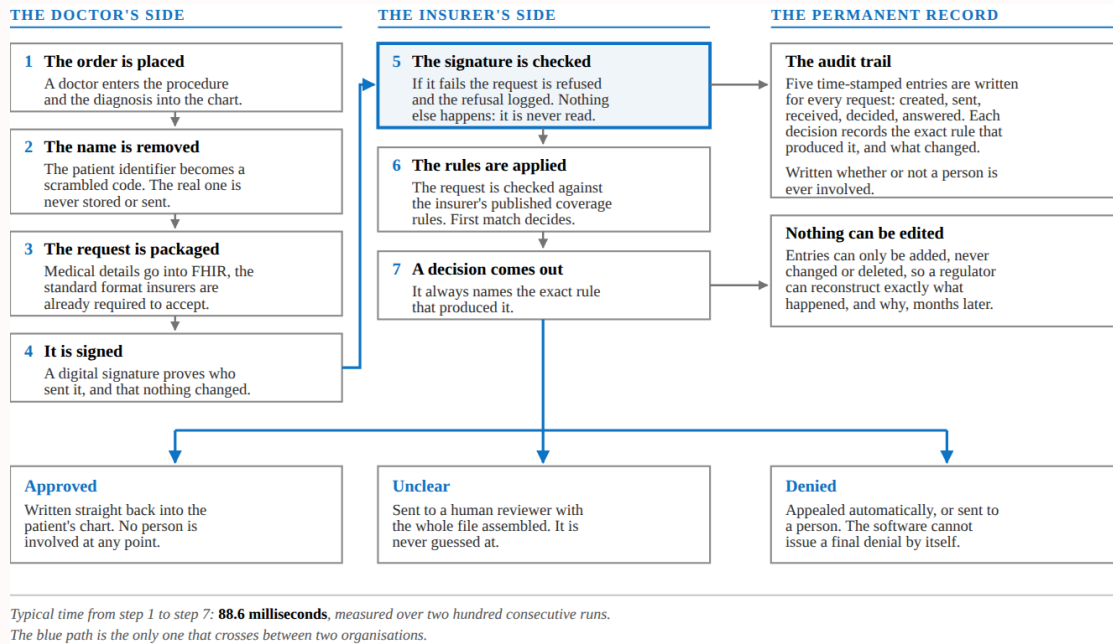
Unclear cases escalate, assembled

A human reviewer opens a file that is already complete: records, rule, signature check, confidence score.

The doctor stays in charge

Medical necessity remains a physician's call. Only the administrative burden disappears.

System Architecture



Stack FastAPI · Next.js 14 · Supabase-ready · FHIR R4 · RSA-2048 signing · ZK circuits (circom/snarkjs) · timestamped audit log

Speed Step 1 to step 7 in 88.6ms (median, n=200)

What We Actually Built

4 of 4

phases built and running

148

automated tests, all passing

18

commits in 11 reviewed batches

22,739

lines added across 260 files

88.6 ms

end-to-end decision

13 days

first commit to working system

```
$ git log --first-parent --pretty="%h %ad %s"
```

```
0a24def 2026-09-19 Document the setting that
                        keeps the site private (#11)
bb02bc4 2026-09-19 Give the site a real mark (#12)
cd0891a 2026-09-19 Make a broken deployment
                        say so (#10)
ee9ed87 2026-09-18 Fix the request detail page (#9)
84c4ca5 2026-09-18 Turn the demo into a website (#8)
8b23760 2026-09-18 Make the demo deployable (#6)
```

```
$ git log --shortstat
```

```
18 commits · 260 file changes
+22,739 insertions · -1,215 deletions
```

Reviewed and merged in eleven batches, 7 to 19 September 2026.

The System Running

Not mock-ups. Both screens are the same build that would go to a customer.

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END-TO-END DEMO

One order, start to finish

Every panel below is live data from the running system — the same agents, rules, signatures, and circuit the rest of the app uses.

Scenario

MRI brain — approves **Knee replacement, matching Dx — approves** Knee replacement, other Dx — escalates Office visit — approves

Unknown procedure — escalates

Pavo-R862 The specific knee rule matches first and approves.

Procedure (CPT) **27447** Diagnosis (ICD-10) **M17.11** **Run Full Demo** Next completed in 11.0s

EHR trigger A physician places an order.

Identity Both agents prove who they are.

Zero-knowledge Criteria proven, PHI withheld.

Rule engine A deterministic rule decides.

Decision The outcome and its audit trail.

Insure What the patient would actually pay.

STEP 6 — INSURE

Arthroplasty, knee, condyle and plateau (total knee replacement) CPT 27447 - deductible not met

The same authorization now drives a price comparison for the member — what they would actually pay, at each facility, through insurance or in cash.

#1	#2	#3
Northgate Community Hospital	Parkside Diagnostic Clinic	Riverside Imaging & Surgical
\$5,296	\$5,977	\$6,279
negotiated \$26,482 - cash \$7,176	negotiated \$29,887 - cash \$11,051	negotiated \$31,394 - cash \$11,444
Insurance cheaper	Insurance cheaper	Insurance cheaper

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Authorization requests

Every decision below records the rule that produced it. Counters refresh every 10 seconds.

Authorizations processed 9 7 auto-approved	Average resolution 43 ms created to resolved	Appeals win rate — none decided yet	2K proofs generated 2	ARIA messages 18 signed and verified
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Filter: **All** Pending Approved Escalated Denied Appealed

STATUS	PROCEDURE	DIAGNOSIS	RULE	CONFIDENCE	SUBMITTED	
Approved	27447	M17.11	PAVO-R862	100%	9/20/2026, 2:32:42 AM	View trail
Approved	27447	M17.11	PAVO-R862	100%	9/20/2026, 2:31:59 AM	View trail
Escalated	27447	H25.561	PAVO-R863	100%	9/20/2026, 2:29:42 AM	View trail
Escalated	27447	H54.5	PAVO-R863	100%	9/20/2026, 2:29:41 AM	View trail
Approved	27447	M17.11	PAVO-R862	100%	9/20/2026, 2:29:41 AM	View trail
Approved	78553	R51.9	PAVO-R861	100%	9/20/2026, 2:29:41 AM	View trail
Approved	99214	386.9	PAVO-R864	100%	9/20/2026, 2:29:40 AM	View trail
Approved	78553	G43.999	PAVO-R861	100%	9/20/2026, 2:29:40 AM	View trail
Approved	27447	M17.11	PAVO-R862	100%	9/20/2026, 2:29:40 AM	View trail

Live ARIA feed

Last 10 messages

- AUTH_RESPONSE: 02:32:42 verified request 28252724
- AUTH_REQUEST: 02:32:42 verified request 28252724
- AUTH_RESPONSE: 02:31:59 verified request B0f6b386
- AUTH_REQUEST: 02:31:59 verified request B0f6b386
- AUTH_RESPONSE: 02:29:42 verified request Bcd39c27
- AUTH_REQUEST: 02:29:42 verified request Bcd39c27
- AUTH_RESPONSE: 02:29:41 verified request 256c8c02
- AUTH_REQUEST: 02:29:41 verified request 256c8c02
- AUTH_RESPONSE: 02:29:41 verified request fcd31fa3
- AUTH_REQUEST: 02:29:41 verified request fcd31fa3

Left: all six steps, order through price, run with no further input and finish in 11.0 seconds. **Right:** every row carries the rule that produced it. The two amber rows are knee replacements whose diagnosis did not match the covered condition, and both wait for a human reviewer with the file already assembled.

Customer Discovery and Validation

12+

interviews with hospital administrators and clinical staff

8

insurance provider conversations completed

3

test surveys distributed to prior authorization teams

2

letters of interest received from health systems

100%

of interviewed providers cited prior authorization delays as a top operational pain point

The Safe Outcome Is the Default

Fixed rules · decides

Same input, same answer, every time, and every answer names the exact rule behind it.

Identity checks · coverage rules · price arithmetic · denial-reason sorting

AI models · assist only

Every result carries a confidence score. Below the threshold it goes to a person.

Reading insurance cards · mapping plain English to a medical code · drafting appeal letters

Unmatched request goes to a human reviewer, within 4 hours.

An appeal below 70% confidence is held for a person.

AI provider offline: authorization itself is unaffected.

No AI language model ever decides an authorization, and the software cannot issue a final denial to a patient at all.

LIMITATIONS

What We Have Not Solved

These are real limitations of a prototype. We would rather state them than be caught by them.

Gap	What it needs
The privacy proof reveals which covered condition a patient has	A more advanced circuit design. Scoped for the next phase.
The proof's setup was done on one machine	A proper multi-party ceremony before any real use.
Provider identity is trusted, not verified	One integration against the federal provider registry.
Facility prices are generated, not gathered	The voice agent that calls facilities, plus the price files insurers must now publish.
Five coverage rules, not a rule library	Clinical staff encoding each insurer's criteria: our largest cost, and the real barrier to a competitor.
The learning layer is not built	Deliberate. Until it exists, anything without a definitive rule goes to a person, which is the correct default anyway.

COMPETITION

Everyone Else Still Waits for a Person

Capability	Pavo Cloud	Cohere Health	Availity AuthAI
Software can start a request with no human	✓	–	–
Decision without a human in the loop for clear cases	✓	–	–
Cryptographic sender verification	✓	partial	partial
Zero-knowledge patient privacy proof	✓	–	–

Every competitor still requires a person to start the request. That is a category difference, not a feature gap.

A Market a Regulation Is About to Create

Federal mandate

CMS-0057-F requires FHIR-based prior-authorization APIs by 1 January 2027. Every payer has a hard deadline, and we are already built against the standard.

AI inflection

Agents now need a rail to initiate authorizations on a patient's behalf. Pavo is that rail.

Scale already exists

47 million payer-provider interactions a year still move through portals and fax.

A compliance deadline every payer must meet is a distribution event, not just a regulation.

How Pavo Makes Money

\$3.00

per request, charged to the insurer

\$400

per month, per medical practice

\$20,000

per year, per insurer connection

Source	2026	2027	2028
Medical practices served (average)	55	330	1,400
Requests processed	171,600	1,647,360	8,736,000
Fees per request	\$514,800	\$4,942,080	\$26,208,000
Practice subscriptions	\$264,000	\$1,584,000	\$6,720,000
Insurer connection fees	\$220,000	\$1,680,000	\$3,200,000
Total revenue	\$998,800	\$8,206,080	\$36,128,000

The \$3.00 fee is 85–92% below the \$15–\$40 an insurer spends handling a request by hand today, which is what makes the switch easy to justify.

What It Costs, and What Is Left

31%

gross margin, 2026

82%

gross margin, 2027

87%

gross margin, 2028

Servers are not the expense

Processing one request costs about six hundredths of a cent, because the deciding path uses no AI models at all.

Clinical staff are

Translating each insurer's published rules into the fixed rules the software applies is the largest line in every year, and the reason margins improve with scale rather than with technology.

Year 1 is 31%, not break-even. Early customers need a great deal of hand-holding, and we would rather say so.

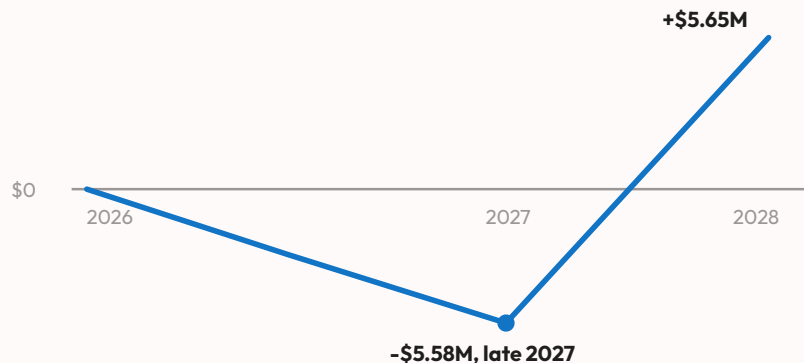
Cost to Win a Customer, and What One Is Worth

Acquisition cost is what it costs in sales and marketing to sign one customer. Lifetime value is the profit that customer produces before they leave, stated over three years.

Measure	Practice	Insurer	Blended
Cost to acquire one customer	\$6,000	\$22,000	\$6,699
Value over three years	\$58,248	\$51,057	\$57,533
Value per \$1 spent acquiring	\$9.70	\$2.30	\$8.60
Months to earn the cost back	3.5	15.2	4.0

Read the insurer column honestly. At \$2.30 back per \$1 spent and fifteen months to recover it, insurer contracts do not pay for themselves as a product. We fund them anyway, because signing one insurer makes Pavo available to every practice that already submits to it. It is a distribution channel, not a profit centre.

We Are Asking For \$8.0 Million



Cumulative cash. The raise covers the trough with roughly twelve months of cushion past breaking even.

The clinical rule library

Our largest cost and the real barrier to a competitor.

Security certification

No insurer signs without it.

Closing the gaps we listed

Provider registry, multi-party ceremony, rule library.

It is not a shopping list. The three largest spending lines across the plan, \$7.46M of engineering, \$5.67M of insurer business development and \$1.72M of clinical staff encoding coverage rules, come to more than the raise on their own, and most of that is paid for out of revenue as it arrives.

If all four of our main assumptions are wrong at once, 2028 revenue is \$13.8 million rather than \$36.1 million: a smaller company, but still a real one.